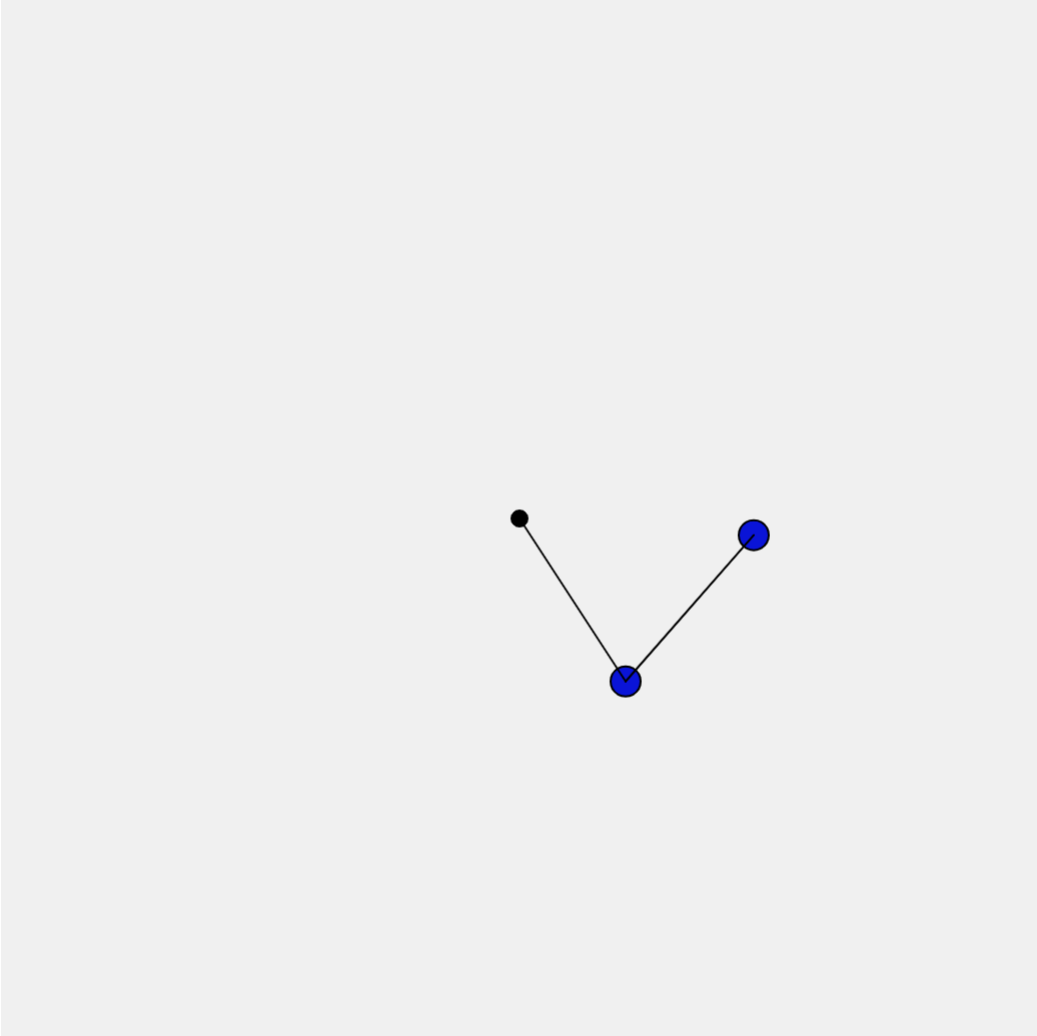
Hello,
My name is Joe.

12th grade student athlete at R.E. Mountain.

3 Body Problem Video: Here



1.1 Double Pendulum

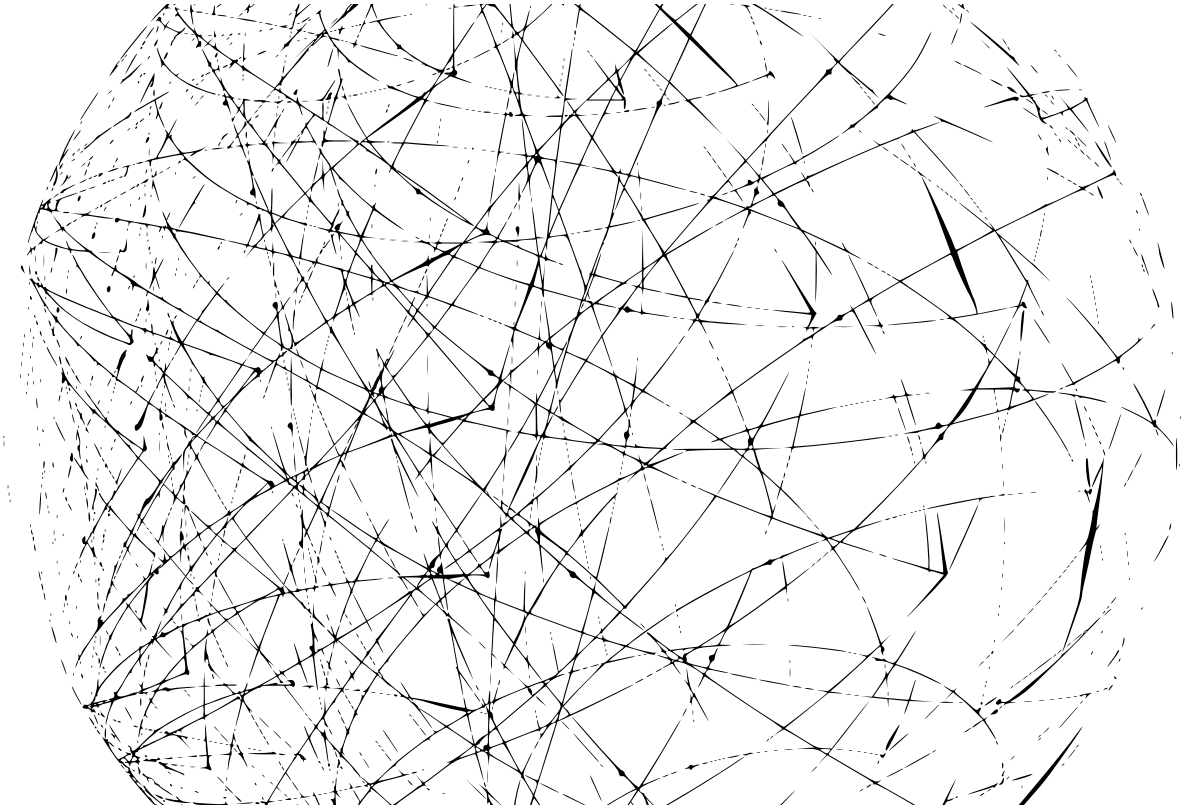
In this project, I find periodic trajectories of a double pendulum through calculations of the Lyapunov exponent at different drop angles. The result is a fractal-like heatmap that outlines the chaotic nature of the double pendulum.

EXPLORE WEB VERSION →

1.2 Debate Motions Network

Debate is a field that interacts with a diverse range of topics. This project aims to use word embeddings to find cosine similarities and ultimately create an organization of debate motions in a network form, allowing for the ability of finding connections between motions that is not possible with the conventional folder structure of organization.

EXPLORE NETWORK →



1.3 Periodic Three Bodies

The three-body problem is known for its chaotic nature. In this project, I developed a new algorithm to measure the periodicity of a system, finding unnamed periodic trajecotires in the 3 body system.

EXPLORE →

2 Athletics

Sports are another passion I enjoy.

2.1 Badminton

Badminton is a sport I have enjoyed since I was very little. Playing this sport has lead me to meeting many new people that I enjoy spending time with.

HIGHLIGHTS →

2.2 Basketball

Basketball is a sport I love to compete in, getting the opportunity to connect to different worlds as meet new people and see great things on travel tournaments.

HIGHLIGHTS →

Contact

I am always excited to connect with others who share my passions.